

Simboni Misiati Peter

Senior Software Engineer — Payments, Mobile Money & Systems for Low-Connectivity Field Work

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Location: Nairobi, Kenya — EAT / UTC+3. Full overlap with East Africa and with US-Eastern mornings; available for 8am ET meetings. · **English:** professional working proficiency (C1) · **Swahili:** native

PROFILE

Full-stack engineer, 10+ years and 100+ delivered systems, working from Nairobi. Most of what I build moves money or records that somebody depends on: M-Pesa collections and receipting, double-entry ledgers, school fee accounts, microfinance and ERP tooling. I treat money as integer cents and assert it in tests, because a rounding error in a ledger is not a cosmetic bug. I design for the conditions the software actually meets — intermittent connectivity, shared servers, staff on a phone rather than a laptop — and I ship end to end, from schema and API through deployment, monitoring and the runbook the operator reads at one in the morning. MSc in Software Engineering. I have built almost exclusively for Kenyan organisations: rural schools, small traders, NGOs and community programmes.

WHAT I WOULD BRING TO THIS ROLE

- **Mobile money in production, in Kenya.** M-Pesa Daraja STK Push collection with automatic receipting, per-vendor credentials encrypted at rest, and Kopo Kopo subscription billing — the same rails GiveDirectly uses to reach recipients.
- **Correctness-first money handling.** Integer KES cents, VAT in basis points, double-entry ledgers recording every transaction for reconciliation, with the arithmetic unit-tested rather than assumed.
- **Systems that keep working when the network does not.** HMAC-signed tickets that verify offline; offline-first PWAs with sync on reconnect; database constraints that make an action idempotent so a retry on a bad line cannot double-pay or double-redeem.
- **Deployment that cannot take the neighbours down.** My most recent release joined a new service to a server already running a live point-of-sale and nineteen other containers: config validated inside the running proxy, applied by graceful reload, every existing site probed before and after, automatic rollback if any of them regressed.
- **Security by construction.** Fail-closed PostgreSQL Row-Level Security with the application role set *NOBYPASSRLS*, two-stage scoped JWTs, Argon2id, append-only audit logs — and a CI gate that asserts cross-tenant isolation on every push, so the guarantee is tested rather than documented.
- **Proximity to the people served.** My clients are a rural school in Bungoma County, a chemicals trader, community organisations and NGOs. I have spent ten years building for users on cheap Android phones and patchy data, which is the environment, not an edge case.

TECHNICAL SKILLS

Backend & APIs	NestJS 11, Node.js, Laravel 12 / PHP 8.4, Python / Django, REST APIs, raw SQL, server actions, cron & background jobs
Languages	TypeScript (strict), PHP 8.4, Python, JavaScript, SQL — object-oriented service design with dependency injection (NestJS, Laravel)
Databases	PostgreSQL 16 (Row-Level Security, migrations, query tuning), MySQL, SQLite / Cloudflare D1, Redis, Prisma & Drizzle ORM, Supabase
Payments	M-Pesa Daraja STK Push, Kopo Kopo, subscriptions & auto-receipting, double-entry ledgers, VAT/tax logic, reconciliation
Cloud & DevOps	Docker & Compose, Caddy, systemd, Fly.io, Cloudflare Workers/Pages, Railway, Render, Vercel; provisioning scripts, nightly backups, zero-downtime config reload
Testing & CI	Unit-tested domain logic, GitHub Actions, CI security gates, automated accessibility audits (axe / WCAG 2.2) and end-to-end smoke tests with Playwright
Architecture	Multi-tenant SaaS, modular monoliths, serverless (Cloudflare Workers), monorepos with shared types, PWAs, offline-first sync, event/audit trails
Frontend	React 19, Next.js 15/16 (App Router, Server Actions, static export), Tailwind CSS v4, Livewire, responsive and mobile-first UI

SELECTED SYSTEMS

TallyPay — multi-tenant billing & payments SaaS · tallypay.co.ke

Next.js 16 (Server Actions) · Cloudflare Workers / D1 · Drizzle ORM · better-auth · M-Pesa Daraja

- Built the full money cycle — quotation → invoice (deposit, balance, 16% VAT) → receipt — with isolated per-business workspaces and a cross-tenant admin console.
- Correct by design: money as integer KES cents and VAT in basis points, unit-tested; every query scoped through *requireOrg()* so a missing tenant context fails closed.
- M-Pesa STK Push collection with automatic receipting; per-vendor API credentials encrypted at rest; PDF documents and transactional email on a serverless runtime.

64 Theatre — ticketing platform with offline gate scanning · tickets.64theatre.art

Laravel 12 · PHP 8.4 · PostgreSQL 16 · Redis 7 · Livewire · Filament · M-Pesa Daraja

- Online sales with M-Pesa STK checkout and SMS/email delivery, on a modular Laravel monolith.
- **Offline-first gate scanner:** HMAC-signed QR tickets verify with no internet at the door; database constraints guarantee each ticket redeems exactly once even under concurrent scans.
- Double-entry money ledger recording every sale, so takings reconcile against the gate.

StackUp — multi-tenant work-management SaaS (own product) · stackup.co.ke · github.com/simboni/smp-planning

NestJS 11 · PostgreSQL 16 · Next.js 15 · React 19 · TypeScript · Capacitor · Fly.io

- Fail-closed multi-tenancy: the application role is *NOBYPASSRLS*, every workspace table *FORCES* Row-Level Security, and a missing workspace context denies every query.
- Argon2id passwords, two-stage JWTs (identity → workspace-scoped access), rotating refresh tokens, append-only audit log — with a **CI gate asserting cross-workspace isolation on every push**.
- One pnpm monorepo with shared types serving three clients: REST API, web app and a Capacitor Android build.

School management platform — for a rural school in Bungoma County

Next.js 16 · Prisma · PostgreSQL 16 · Docker Compose · Caddy · GitHub Actions

- Fees, invoicing and receipting; attendance; examinations; payroll; transport and boarding; role-based permissions and an audit trail — serving a school of 525 learners.
- **Bulk enrolment from the registers schools actually keep:** tolerant CSV/KNEC import with fuzzy column matching, a dry-run preview showing the outcome of every row before anything is written, and duplicate detection on admission number, national UPI and name-within-class so re-running a file never doubles the roll.
- Deployed onto a shared VPS alongside a live point-of-sale and nineteen other containers, with a release script that validates configuration inside the running reverse proxy, applies it by graceful reload, probes every co-tenant site before and after, and rolls itself back automatically on any regression.
- CI runs an automated accessibility audit (axe, WCAG 2.1/2.2 AA) over every page in both themes at two viewport widths; it has caught functional defects, not only accessibility ones.

Naveedex — trading-journal SaaS · naveedex.com | **Riziki POS** — point of sale for a chemicals trader

Next.js · TypeScript · Supabase · Vercel · Kopo Kopo | Next.js · SQLite · Docker

- Authentication, subscription payments, scheduled jobs, admin tooling, analytics and one-click PDF/Excel export; in-shop point-of-sale running in production daily.

EXPERIENCE

Full-Stack Developer — Freelance & Contract

2015 – Present

Independent — clients across Kenya and remote

- Delivered 100+ projects for 20+ clients: SaaS products, fintech and payment systems, ERP and microfinance platforms, ticketing, e-learning and web applications.
- Own the entire lifecycle: requirements and data modelling, API and schema design, implementation, testing, deployment, monitoring, documentation and handover.
- Integrate third-party systems end to end — mobile money (M-Pesa Daraja, Kopo Kopo), SMS gateways, transactional email, PDF generation, scheduled jobs.
- Work directly with non-technical operators — shopkeepers, school bursars, event staff — translating how they actually work into software they can run without me.

Founder & Senior Developer — SMP Developers

2023 – Present

Nairobi, Kenya

- Lead architecture and development of web platforms, systems and mobile apps; coordinate collaborators and review their code to hold quality and schedule.
- Build StackUp as an in-house SaaS product alongside client engagements.

Chairman & Director — SMP Eventures Ltd

2015 – Present

- Lead a multi-disciplinary team turning client and internal product ideas into delivered software.

EDUCATION

MSc, Software Engineering

Jomo Kenyatta University of Agriculture & Technology
(JKUAT) · 2023 – 2025

Programming Certificate — Software & App Development

JKUAT · Git, JavaScript, iOS, Android, PHP, Laravel, Python, Django · 2024 – Present

Availability and location. Based in Nairobi (EAT / UTC+3) with reliable high-speed internet — a full working-day overlap with East Africa and with US-Eastern mornings, including regular 8am ET meetings. Able to travel for team retreats and field visits. Comfortable owning features end to end from a written brief, and equally comfortable saying when a brief is wrong.